

Builder Basics — Block Placement Practice (Extension · Intermediate)

Topic: Move, Turn, Climb With Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Intermediate · **Time:** about 38 minutes

✖ This is a bonus challenge, not a main lesson. Try it once you can move the agent around, before you start placing blocks by hand.

In the Agent Movement lesson the agent **walked**. Here it walks and **draws** at the same time. You flip one switch — **place on move** — at the very top, and you never touch it again. After that, every step drops a block.

Your commands: `move forward`, `turn left`, `turn right`, `move up`.

🧱 One step = one block. A turn is not a step.

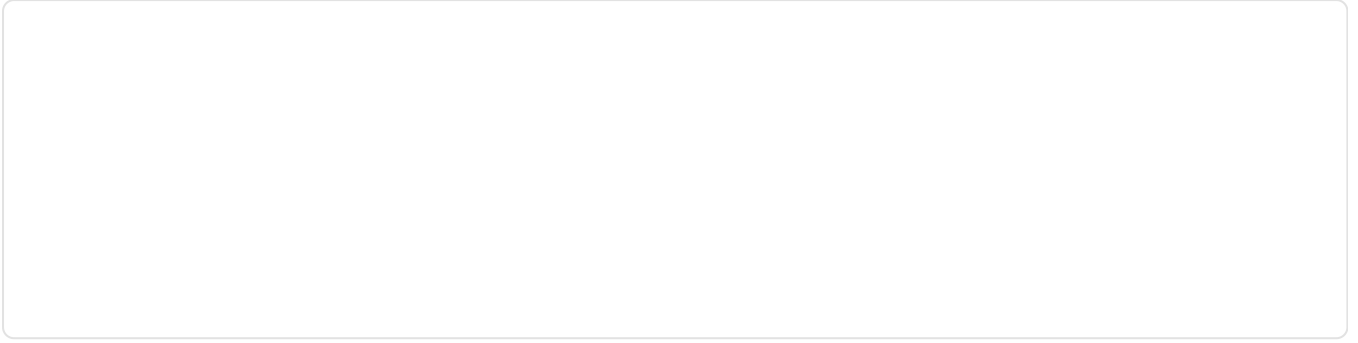
1 · Predict 🧠

The switch is ON at the top and stays ON.

```
place on move ON
set block to stone
move forward by 3
turn right
move forward by 2
```

How many blocks land in total? Show your adding.

Draw the shape those blocks make. Mark where the agent started with an S.

A large, empty rectangular box with a thin black border, intended for drawing the shape of blocks and marking the starting point of an agent.

2 · A Turn Is Not a Step

- `move forward by 2` — the agent **travels** two blocks. Two steps, so two blocks drop.
- `turn right` — the agent **spins** on the spot. It goes nowhere. No step, so **no block**.

The switch is ON. The agent runs `turn left`. How many blocks does it place? Circle one: 0 · 1 · 2

Why? Explain in one sentence.

Both of these programs place 4 blocks. Explain what is different about the two trails.

Program A	Program B
<code>move forward by 2</code>	<code>move forward by 2</code>
<code>move forward by 2</code>	<code>turn right</code>
	<code>move forward by 2</code>

3 · Count the Blocks 1234

Follow this program one line at a time. Fill in the table. Turns get a row too.

```
place on move ON
set block to stone
move forward by 4
turn right
move forward by 3
turn right
move forward by 4
```

Command	Blocks this line	Total so far
move forward by 4		
turn right		
move forward by 3		
turn right		
move forward by 4		

Two lines placed zero blocks. Which ones, and why?

4 · Going Up

`move up` is a step too, so it drops blocks as well. Up steps stack blocks instead of laying them flat. Mix `move forward` and `move up` and you get **stairs**:

```
place on move ON
set block to stone
move forward by 1
move up by 1
move forward by 1
move up by 1
move forward by 1
```

How many blocks does this staircase use?

Draw this staircase from the side. Mark the start with an S.

5 · Spot the Bug 🐛

Bug A — This should draw a line 5 blocks long.

```
set block to stone  
move forward by 5
```

What is missing from the program? What is on the ground when it finishes?

Bug B — This should draw a corner: 3 blocks, turn, 3 more blocks.

```
place on move ON  
set block to stone  
move forward by 3  
turn right  
turn right  
move forward by 3
```

The agent placed 6 blocks, but there is no corner. Explain what those two turns did.

6 · Design Your Own Path

Plan a trail before you build it. Use **move forward** , **turn left** , **turn right** and **move up** . The switch goes ON once and stays ON.

Choose your block and your path. Fill in the blanks:

My block is ...

move forward by ... (... blocks)

... (... blocks)

... (... blocks)

How many blocks does your path place in total?

Draw the shape your path will make. Mark the start with an S and the end with an E.

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7 • Show Your Work 📷 🎥

Open your world and build the path you designed in Section 6. Flip **place on move** ON once at the top, then never touch it again. If the shape comes out wrong, change your numbers and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.